Process Characterization Report

A-Mab Drug Substance — Anion Exchange Chromatography (Step 8) — PCR-008

Process Development / Manufacturing Science & Technology (MSAT)

2026-07-24

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Field	Value
Document ID	PCR-008
Document class	Process Characterization Report
Title	Process Characterization Report: Anion Exchange Chromatography (Step 8)
Product	A-Mab
Version	1.0
Effective date	2026-07-24
Status	Approved (synthetic)
Document owner	Manufacturing Science & Technology (MSAT)
Sponsor / sites	Novacyte Biologics; Novacyte Biologics — Cambridge, MA (Development) → Novacyte Biologics — Grafton, WI (Commercial DS)

Approvals

Table 1: Approval record (synthetic; signatures not applied).

Role	Function	Name (synthetic)	Signature	Date
Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
Reviewer	Manufacturing Science	—	—	—
Reviewer	Statistics	—	—	—
Approver	Quality Assurance	—	—	—

Abbreviations

ADCC, antibody dependent cellular cytotoxicity. AEX, anion exchange chromatography. AMV, analytical method validation. BLA, Biologics License Application. CCD, central composite design. CEX, cation exchange chromatography. CPP, critical process parameter. CQA, critical quality attribute. Cpk, one sided process capability index. CV, coefficient of variation. DNA, deoxyribonucleic acid. DoE, design of experiments. DS, drug substance. ELISA, enzyme linked immunosorbent assay. GPP, general process parameter. HCP, host cell protein. ICH, International Council for Harmonisation. icIEF, imaged capillary isoelectric focusing. IgG1, immunoglobulin G subclass. KPP, key process parameter. LRF, log reduction factor. MSAT, Manufacturing Science and Technology. MVM, minute virus of mice. NOR, normal operating range. OLS, ordinary least squares. PAR, proven acceptable range. PPQ, process performance qualification. QbD, quality by design. RMSE, root mean square error. RSM, response surface methodology. SD, standard deviation. SDM, scale-down model. SOP, standard operating procedure. TCID50, median tissue culture infectious dose. UV, ultraviolet absorbance. WC-CPP, well controlled critical process parameter. XMuLV, xenotropic murine leukaemia virus. qPCR, quantitative polymerase chain reaction.

Executive summary

Anion exchange chromatography is the last chromatographic step in the A-Mab drug substance process. It is operated in flow-through mode. At the load pH the antibody carries a net positive charge and passes through the quaternary amine ligand, while species that are negatively charged at that pH bind. The step therefore removes host cell protein, residual DNA, leached Protein A and virus particles without eluting the product. This report presents the Stage 1 characterization of the step, executed against the plan PCP-008 on a qualified scale-down model of the commercial process, which is fed from a 15,000 L production bioreactor. The scope of the study was set by the pre-characterization risk assessment RA-001.

Of the 5 parameters carried into the study, 4 were studied in a multivariate design and 1 was assessed one parameter at a time. A two-level full factorial screening design of 19 runs was followed by a face-centred central composite design of 28 runs in the same 4 factors. Every run was assayed for 4 responses. The study recorded 3 deviations, which are reported in §13. One of them invalidated the first execution of both designs, which were re-executed in full on a requalified load. The analysis reported here is the re-executed study.

Load pH and the equilibration and wash conductivity govern host cell protein clearance, and they act on it in opposite directions. Load pH and load conductivity govern the clearance of both model viruses. Protein load carried no effect on any response over the range studied, and the univariate assessment of operating flow rate identified none over its range. The response-surface models for pool HCP, XMuLV LRF and MVM LRF explain between 0.90 and 0.93 of the variance in the data, with no significant lack of fit (the smallest lack-of-fit p-value is 0.50). Step yield is high in flow-through mode and no parameter affected it. Over the characterized region, 86 % of the parameter combinations meet every criterion the step governs. Pool HCP is the response that binds the region and the viral responses reject none of it. The normal operating ranges are not fully covered. At the least favourable point inside them the model predicts pool HCP of 22.3 ng/mg against an in-process limit of 21.7 ng/mg, and §7, §8 and §10 report what follows for the control strategy.

The step is the sole contributor to the MVM clearance claim other than virus filtration. It delivers 4.71 log10 of the cumulative 10.03 log10 against a requirement of 8.6 log10, and 5.95 log10 of the cumulative enveloped-virus clearance. Cumulative MVM clearance carries a one-sided Cpk of 1.51, which is the tightest capability of the drug substance. All 5 parameters were classified as well controlled critical process parameters. The step contributes parameter ranges, an in-process limit on pool host cell protein, a modular viral clearance claim and a control on the hold time of the incoming cation exchange pool to the control strategy. These outcomes roll up into PCMR-001.

1 Introduction

1.1 Product and unit operation

A-Mab is a humanized IgG1 monoclonal antibody. Its primary mechanism of action is antibody dependent cellular cytotoxicity, and the glycan attributes that modulate that mechanism are formed in the production bioreactor and are not changed downstream. The drug substance process is a platform process. Protein A capture is followed by low-pH viral inactivation, cation exchange chromatography, anion exchange chromatography, small virus retentive filtration and ultrafiltration with diafiltration.

Anion exchange chromatography is Step 8 of that train. The step is operated in flow-through mode on a strong anion exchange resin carrying a quaternary amine ligand. The column is equilibrated, the cation exchange pool is adjusted and loaded, and the product is collected in the flow-through and the first wash. The antibody has a basic isoelectric point and is positively charged at the load pH, so it does not bind. Host cell proteins that are acidic at the load pH, residual DNA, leached Protein A and virus particles carry a net negative charge at that pH and bind the ligand.

The step forms no quality attribute. It sets the cumulative clearance of minute virus of mice, and it is a clearance step for enveloped virus, host cell protein, residual DNA and leached Protein A. In the nominal process the step reduces host cell protein 6.2-fold to 18.9 ng/mg and recovers 97.3% of the product mass. Viral clearance is claimed for the step and the basis for that claim is given in §8.

1.2 Objectives

The objectives of the study were the following.

(i) Quantify the relationship between each process parameter and the quality attributes and process performance attributes the step governs, over ranges wider than the intended routine control ranges.
(ii) Define the operating region within which the step meets every criterion it is responsible for.
(iii) Establish a proven acceptable range for each parameter against each attribute it governs, and determine where the normal operating ranges sit inside them.
(iv) Classify each parameter according to its demonstrated impact and the risk that it leaves the operating region, per SOP-4001.
(v) Justify the ranges to be verified in Stage 2 and state what this step contributes to the control strategy.

1.3 Regulatory and scientific basis

The study is a Stage 1 process design activity in the lifecycle approach to process validation (U.S. Food and Drug Administration 2011). The enhanced approach of ICH Q8, Q9 and Q11 was applied, with prior knowledge and a documented risk assessment used to set the study scope and with criticality treated as a continuum rather than as a binary classification (International Council for Harmonisation 2009, 2023b, 2012). Viral clearance is evaluated as a modular claim under ICH Q5A(R2), which requires each contributing step to be studied independently on a qualified small scale spiking model (International Council for Harmonisation 2023a). The process model, the quality attribute framework and the risk ranking method follow the A-Mab case study (CMC Biotech Working Group 2009).

1.4 Related documents

The characterization package this report belongs to is listed in Table 1.1. RA-001 scoped the study, PCP-008 specified it, and this report records what was executed and what it established.

Table 1.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCP-008	Process Characterization Plan (Anion Exchange Chromatography (Step 8))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

2 Prior knowledge and quality risk basis

2.1 Platform and prior-product knowledge

The anion exchange polishing step has been operated in flow-through mode for three previously licensed antibodies produced on the same platform (X-Mab, Y-Mab and Z-Mab) and throughout A-Mab clinical manufacture. The resin, the buffer system and the mode of operation are unchanged for A-Mab. This experience establishes the mechanism of separation and the parameters that control it, so characterization confirms and bounds the platform behaviour rather than discovering it.

The separation rests on a single equilibrium. An impurity binds the quaternary amine ligand according to its net charge at the load pH and according to how strongly the ionic strength of the buffer shields that charge. Load pH and conductivity are therefore the two parameters expected to set the clearance of every bound species, and the antibody is expected to pass through unretained across the whole range because its isoelectric point lies well above the load pH.

The expected behaviour of each attribute follows from that equilibrium. Host cell proteins are a population with a distribution of isoelectric points. Raising the load pH moves more of that population below its isoelectric point and into the bound fraction, and raising the conductivity of the equilibration and wash buffer shields the interaction and allows the more weakly bound species through with the product. DNA is a polyanion at any pH the step could operate at, binds strongly, and is expected to be cleared by orders of magnitude with little dependence on the operating parameters. Leached Protein A is acidic and binds while the antibody passes through, except where it remains complexed with the antibody through the Fc region. Retrovirus-like particles and the parvovirus model both carry acidic surfaces and bind by the same electrostatic mechanism, so their clearance is expected to fall as load pH drops and as conductivity rises. Yield in flow-through mode is expected to be high and to be insensitive to the parameters, because loss can only occur if the antibody's charge approaches neutrality at the load pH.

Prior knowledge does not establish where within the proposed ranges the clearance of each species becomes limiting, and it does not establish whether the parameters interact. Those two questions are what the designed experiments in §4 were run to answer.

2.2 Quality attributes in scope

Table 2.1 gives the attribute this step sets. Criticality was assigned by the impact and uncertainty assessment of the A-Mab framework, in which an attribute scored high or very high is treated as critical (CMC Biotech Working Group 2009).

Table 2.1: Quality attribute set by the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — MVM (parvovirus)	viral safety	8.6–20 log10 (cumulative)	VH	20

Clearance of minute virus of mice is a very high criticality attribute and the acceptance criterion is cumulative across the process. The step is judged against the log reduction it must deliver for the cumulative requirement to be met, which is derived in §7.

Table 2.2 gives the attributes the step clears but does not set. Each is formed upstream and is brought to its drug substance criterion by more than one step.

Table 2.2: Quality attributes cleared by the anion exchange step.

CQA	Category	Acceptance	Criticality	Tool #1
Viral clearance — XMuLV (enveloped)	viral safety	16.7–30 log10 (cumulative)	VH	20
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Charge variants and the glycan attributes are not affected by charge-based separation at the levels present in A-Mab and are not measured across this step. They are formed in the production bioreactor and are reported in PCR-003. Aggregate is reduced principally by cation exchange and is reported in PCR-007.

2.3 Risk-based prioritization and parameter selection

RA-001 ranked every parameter of the step on its potential impact on a quality attribute and on its potential to interact with other parameters, and assigned each one a study type. 4 parameters were assigned to a multivariate design because they act on the same binding equilibrium and were expected to interact. Operating flow rate was assigned to a univariate assessment because it acts on residence time rather than on the equilibrium, and the binding of the small, highly charged species this step removes is fast relative to the residence time the range spans. Parameters held to platform specification, such as the buffer species and the resin, are identity and quality controlled rather than characterized.

The ranges taken forward were widened well beyond the intended routine control ranges, so that the study bounds the knowledge space rather than the control space. Table 4.1 in §4 gives both. The widening is deliberate and it is what allows an excursion to be assessed against data rather than against an extrapolation of the model.

3 Materials and methods

3.1 Scale-down model and its qualification

The studies were executed on a laboratory scale chromatography system qualified as a model of the commercial step under SOP-1001. The model was designed on established principles of chromatographic scaling. Bed height, linear flow velocity, protein load per volume of resin, and the equilibration, load, wash and strip volumes normalized to column volumes were held equal to the commercial values. Column efficiency was verified by plate count and peak asymmetry before each campaign, and the resin was drawn from the same lot family used at scale and operated within the cycle number limits of SOP-2008.

Qualification compared the model with at-scale performance on the input and output attributes that matter to this step, which are the host cell protein, residual DNA and leached Protein A concentrations of the load and of the pool, and the step yield. The comparison was made on replicate runs at the set-point conditions. No difference in column efficiency or in step performance between scales was identified, and the model is therefore suitable for the studies reported here.

Viral clearance cannot be measured at manufacturing scale, because live virus may not be introduced into a manufacturing facility. A separate small scale spiking model was qualified for the viral runs under ICH Q5A(R2) (International Council for Harmonisation 2023a). The spiking model uses the same column geometry, the same resin lot family and the same buffers as the process model, with the load spiked with XMuLV or MVM before adjustment. Spike volume was held low enough that the load composition was not materially altered, and unspiked controls were run alongside to confirm that the spike did not change the host cell protein or yield performance of the step.

The centre-point replicates of each design are the reproducibility measurement of the model under the conditions of the study. They provide the pure-error term used in the lack-of-fit tests in §5.3, and their spread is reported in §5.1.

3.2 Operation

Each run followed SOP-2011. The column was sanitized and equilibrated with the equilibration buffer to the conductivity given by the run setting. The cation exchange pool was adjusted to the load pH and load conductivity of the run and loaded to the protein load of the run at the operating flow rate. The product was collected in the flow-through and the first wash, and collection was stopped on the trailing edge of the ultraviolet absorbance trace. Deviation DEV-008-02 concerns that collection set-point and is reported in §13.2. The column was stripped and sanitized after each run and was not re-used across runs within a design block.

The load for every run in the reported execution was drawn from a single requalified cation exchange pool lot, so that load composition is a constant of the study rather than a source of variance. Load charge variant content was verified by icIEF before the study began. The reason that verification exists is DEV-008-01, which is reported in §13.1.

3.3 Analytical methods

Table 3.1 lists the controlled procedures and validated methods used. Pool host cell protein was measured by immunoassay under AMV-3012, residual DNA by quantitative PCR under AMV-3014 and leached Protein A by immunoassay under AMV-3016. XMuLV and MVM titres were measured under AMV-3017 and AMV-3018 and are reported as the log10 reduction factor across the step. Charge variants of the load were measured by icIEF under AMV-3013. Step yield was calculated from the product mass in the pool relative to the product mass loaded. Method performance is summarized in Appendix D.

Table 3.1: Controlled procedures and validated analytical methods.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2011	Operation of the Anion-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation
AMV-3017	XMuLV Infectivity Titre (TCID50)	Method validation
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation

The variance of the host cell protein immunoassay is the largest analytical contribution to the responses of this study, and it is not separable from process variance in the designs as executed. The centre-point spread reported in §5.1 therefore bounds the sum of the two, and every statement about reproducibility in this report is made against that combined term rather than against a process-only estimate.

3.4 Sampling plan

Load, flow-through pool and strip samples were taken on every run. The pool sample is the mixed pool over the whole collected fraction, so the reported pool concentration is the mass average and not a point value on the trailing edge. Host cell protein, residual DNA and leached Protein A were assayed on the load and the pool of every run. Virus titres were assayed on the spiked load and the pool of every run of the spiking model. The screening design carried 3 centre-point replicates and the response-surface design carried 4 (§4). Assays were run in single determination on process samples, with the assay controls of the method validation applied to each plate.

3.5 Statistical methods

The coded levels −1 and +1 correspond to the low and high edges of each characterization range, and every model was fitted on the coded factors by ordinary least squares. The screening data were fitted with a model carrying all 4 main effects and all 6 two-factor interactions. An effect is reported as twice the coded coefficient, which is the change in the response between the low and high edges of the range. The response-surface data were fitted with the full quadratic model in the 4 factors. Significance was assessed at $\alpha = 0.05$ throughout.

Model adequacy was judged on the coefficient of determination, the adjusted and predicted coefficients of determination, the overall F-test, and an analysis of variance in which the residual sum of squares is partitioned into lack of fit and pure error. The pure-error term comes from the centre-point replicates, so a significant lack of fit means the model misses structure that the replicates show to be real. The predicted coefficient of determination is computed from the prediction error sum of squares, which refits the model with each run removed in turn and is therefore the measure that penalizes an over-fitted model.

A factor with no term significant at $\alpha = 0.05$ in either design is reported as having no demonstrated effect over the range studied. That is a statement about the range and the power of the design and not a statement that the parameter cannot matter, and the classification in §9 treats it accordingly.

Commercial-scale capability was estimated by Monte-Carlo simulation of 2,000 batches with every parameter varying inside its normal operating range, propagated through the fitted models of the whole train, and is reported as a one-sided Cpk against the drug substance criterion (§8).

4 Study design

4.1 Factors, ranges and the knowledge space

Table 4.1 gives the parameters, their set-points, their normal operating ranges and the characterization ranges over which they were studied. The characterization range is the knowledge space edge and is not a proven acceptable range. The proven acceptable ranges are computed per attribute in §7.

Table 4.1: Parameters, ranges, study type and final classification.

Parameter	Unit	Set-point	NOR	Char. range	Class	Study
Load pH	pH	7.5	7.35–7.65	7.2–7.8	WC-CPP	multivariate
Equil/Wash-1 conductivity	mS/cm	2.6	2.1–3.1	1.6–3.6	WC-CPP	multivariate
Load conductivity	mS/cm	5.5	4–7	3–8	WC-CPP	multivariate
Protein load	g/L resin	200	120–280	50–300	WC-CPP	multivariate
Operating flow rate	cm/hr	300	200–400	150–450	WC-CPP	univariate

The characterization range of every parameter is at least twice the width of its normal operating range. Load pH was studied over 7.2 to 7.8 against a normal operating range of 7.35 to 7.65, and the equilibration and wash conductivity over 1.6 to 3.6 mS/cm against a normal operating range of 2.1 to 3.1 mS/cm. Protein load was studied from 50 to 300 g per litre of resin, which reaches well below and somewhat above the routine range, because load is the parameter most likely to expose a capacity limit on the bound species. The upper edge of the load pH range was set by the point at which the antibody itself begins to acquire appreciable negative charge, and the lower edge by the point at which impurity binding weakens.

Table 4.2 gives the coded letter used for each factor in the effect and coefficient tables that follow.

Table 4.2: Coded factor legend.

Code	Factor	Unit	Studied in
A	Load pH	pH	screening + RSM
B	Equil/Wash-1 conductivity	mS/cm	screening + RSM
C	Load conductivity	mS/cm	screening + RSM
D	Protein load	g/L resin	screening + RSM

4.2 Screening design

The screening study was a two-level full factorial in the 4 factors with 16 factorial runs and 3 centre-point replicates, 19 runs in total, executed in randomized order. A full factorial at this size estimates every main effect and every two-factor interaction without aliasing, which was the reason for choosing it over a fractional design. The purpose of the screening study is to identify which parameters carry an effect and which do not. It does not resolve curvature, and the magnitudes it returns are refined by the response-surface study. The design matrix and the measured responses are in Appendix A.

4.3 Response-surface design

The response-surface study was a face-centred central composite design in the same 4 factors, with 16 factorial runs, 8 axial runs at the faces of the cube and 4 centre-point replicates, 28 runs in total. The axial points sit on the faces rather than outside them, so no run leaves the characterization range and the design supports a quadratic term in each factor without extrapolating. All 4 screening factors were carried forward, because each of them either carried a significant effect on at least one response or lay on a mechanistic path that the screening design could not close. The response-surface model is the predictive model of this report, and every prediction, design space statement and proven acceptable range in §6 to §8 is derived from it. The design matrix and the measured responses are in Appendix B.

4.4 Univariate assessment

Operating flow rate was assessed one parameter at a time over 150 to 450 cm/hr rather than in the multivariate design. Flow rate sets the residence time in the bed. Binding of the small, highly charged species this step removes is fast on the timescale the range spans, and there is no mechanism by which residence time would change the position of the binding equilibrium that load pH and conductivity control. An interaction with the multivariate factors was therefore not expected and the parameter was kept out of the design in order to spend the runs on the factors that act on the equilibrium. The parameter remains quality linked through residence time and is classified accordingly in §9.

5 Results

5.1 Centre-point performance and reproducibility

Table 5.1 gives the mean, the standard deviation and the coefficient of variation of each response over the 4 centre-point replicates of the response-surface design. These replicates were run at identical settings, so their spread is the combined process and analytical variability of the model, and it is the pure-error term against which lack of fit is tested in §5.3.

Table 5.1: Centre-point reproducibility, response-surface design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	4	13.7	2.48	18.2
XMuLV LRF ($\log_{10}$)	4	6.63	0.445	6.71
MVM LRF ($\log_{10}$)	4	4.9	0.336	6.86
Step yield	4	0.966	0.0123	1.27

Pool HCP is the least precise response at 18.2 % coefficient of variation, which is consistent with the precision of the immunoassay at the concentrations this step delivers. The two viral responses reproduce at 6.7 % and 6.9 %, which for a log10 quantity is a spread of a few tenths of a log and is normal for an infectivity titration. Step yield reproduces at 1.3 %. The centre points of the screening design gave 20.6 % on pool HCP and 5.6 % on MVM LRF, so the two designs reproduce to the same degree and their results may be read together.

The consequence for the rest of this section is that an effect on pool HCP has to be large to be detected, and that a difference of a few tenths of a log in a viral response is not distinguishable from assay noise. Both bounds are applied where they matter below.

5.2 Screening factor effects

Table 5.2 summarizes the screening models. An effect is the change in the response between the low and the high edge of the characterization range.

Table 5.2: Screening model summary, all responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	19	0.966	0.923	0.841	22.6	8.55e-05	2.45
XMuLV LRF (log_{10})	19	0.842	0.644	-0.106	4.25	0.0257	0.393
MVM LRF (log_{10})	19	0.97	0.932	0.869	25.8	5.18e-05	0.178
Step yield	19	0.498	-0.13	-3.64	0.792	0.642	0.0126

Pool HCP was affected by the equilibration and wash conductivity, by load pH, and by the interaction between them. Table 5.3 gives the six largest effects. Raising the equilibration and wash conductivity across its range raised pool HCP by 12.1 ng/mg ($p < 0.001$) and raising load pH across its range lowered it by 11.9 ng/mg ($p < 0.001$). The interaction between the two is -6.6 ng/mg ($p < 0.001$), which is about half the size of either main effect. Neither load conductivity nor protein load affected pool HCP.

Table 5.3: Screening effects on pool HCP, six largest.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	

Both viral responses behaved the same way as each other and differently from pool HCP. Table 5.4 gives the effects on MVM LRF. Raising load pH across its range raised MVM LRF by 1.22 log10 ($p < 0.001$) and raising load conductivity lowered it by 0.71 log10 ($p < 0.001$). The corresponding effects on XMuLV LRF are 1.00 log10 ($p < 0.001$) and -0.60 log10 ($p = 0.015$). No interaction reached significance for either virus.

Table 5.4: Screening effects on MVM LRF, six largest.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	

The equilibration and wash conductivity, which is the parameter that governs host cell protein clearance, did not reach significance on either virus. Its effect on MVM LRF is 0.18 log10 ($p = 0.076$) and on XMuLV LRF 0.20 log10 ($p = 0.342$). Both are within the reproducibility of the titration reported in §5.1, so the screening study separates the parameter that governs the protein impurity from the parameter that governs virus removal.

No parameter affected step yield. The screening model for yield is not significant overall ($p = 0.642$) and the largest single effect, -0.0116 on a fraction, is smaller than the centre-point standard deviation of the response. This is the expected result for a flow-through step in which the product does not bind.

The screening models for pool HCP and MVM LRF fit the data closely ($R^2 = 0.97$ and 0.97), but the model is nearly saturated at 19 runs and its predicted coefficient of determination for XMuLV LRF is negative (-0.11). The screening study is therefore read here as an identification of which factors matter, and no prediction in this report rests on it.

5.3 Response-surface models

Table 5.5 summarizes the response-surface models. These are the predictive models of the study.

Table 5.5: Response-surface model summary, all responses.

Response	N	R^2	Adj. R^2	Pred. R^2	F	p-value	RMSE
Pool HCP (ng/mg)	28	0.934	0.864	0.66	13.2	1.83e-05	2.65
XMuLV LRF ($\log_{10}$)	28	0.915	0.824	0.621	10	8.69e-05	0.314
MVM LRF ($\log_{10}$)	28	0.898	0.787	0.394	8.14	0.000269	0.32
Step yield	28	0.488	-0.0632	-2.18	0.885	0.589	0.0145

The models for pool HCP, XMuLV LRF and MVM LRF are all significant and explain between 0.90 and 0.93 of the variance in the data. The predicted coefficients of determination remain positive for all three, with the smallest at 0.39 for MVM LRF, so the quadratic model is supported by the data rather than fitted to its noise. The model for step yield is not significant ($p = 0.589$), which repeats the screening result. Step yield maps to no acceptance criterion at this step and is reported as a process performance attribute only.

Table 5.6 gives the full coefficient table for pool HCP and Table 5.7 its analysis of variance. Load pH and the equilibration and wash conductivity carry coefficients of -5.34 and 6.19 ng/mg per coded unit, and their interaction -1.45 ($p = 0.047$). No quadratic term is significant, so the surface is close to planar with a twist. Lack of fit is not significant (p = 0.499), which means the residual scatter is what the centre-point replicates already showed the assay and the process to carry.

Table 5.6: Response-surface coefficients, pool HCP.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	14.8	0.917	16.2	5.51e-10	***
A	-5.34	0.625	-8.54	1.09e-06	***
B	6.19	0.625	9.89	2.04e-07	***
C	0.0399	0.625	0.0637	0.95	
D	0.387	0.625	0.619	0.546	
AB	-1.45	0.663	-2.19	0.0475	*
AC	-0.925	0.663	-1.39	0.187	
AD	0.0192	0.663	0.029	0.977	
BC	0.15	0.663	0.226	0.825	
BD	-0.559	0.663	-0.843	0.415	
CD	-0.124	0.663	-0.187	0.855	
A^2	1.91	1.65	1.16	0.268	
B^2	2.22	1.65	1.34	0.202	
C^2	-0.0187	1.65	-0.0113	0.991	
D^2	-1.98	1.65	-1.2	0.253	

Table 5.7: Analysis of variance with lack of fit, pool HCP.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	1,302.7	14	93.1	13.2	1.83e-05
Residual	91.5	13	7.04	nan	nan
Lack of fit	73.0	10	7.3	1.19	0.499
Pure error	18.5	3	6.16	nan	nan

Table 5.8 and Table 5.9 give the same two tables for MVM LRF, which is the attribute this step sets. Load pH and load conductivity carry coefficients of 0.54 and -0.56 log10 per coded unit, both with $p < 0.001$ and $p < 0.001$. The two are of nearly equal magnitude and opposite sign. No interaction and no quadratic term is significant, and lack of fit is not significant (p = 0.619). The surface for MVM clearance is therefore a plane over the region studied, tilted by load pH one way and by load conductivity the other.

Table 5.8: Response-surface coefficients, MVM LRF.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	4.85	0.111	43.9	1.63e-15	***
A	0.539	0.0754	7.15	7.48e-06	***
B	0.0307	0.0754	0.407	0.691	
C	-0.564	0.0754	-7.48	4.65e-06	***

Term	Coef.	Std. err.	t	p-value	Sig.
D	0.0261	0.0754	0.346	0.735	
AB	0.127	0.08	1.59	0.136	
AC	-0.074	0.08	-0.925	0.372	
AD	0.0353	0.08	0.441	0.666	
BC	-0.0339	0.08	-0.424	0.679	
BD	0.00805	0.08	0.101	0.921	
CD	-0.0156	0.08	-0.195	0.848	
A^2	-0.322	0.199	-1.61	0.13	
B^2	0.149	0.199	0.747	0.469	
C^2	0.0224	0.199	0.112	0.912	
D^2	0.135	0.199	0.679	0.509	

Table 5.9: Analysis of variance with lack of fit, MVM LRF.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	11.7	14	0.834	8.14	0.000269
Residual	1.33	13	0.102	nan	nan
Lack of fit	0.994	10	0.0994	0.881	0.619
Pure error	0.339	3	0.113	nan	nan

XMuLV LRF behaves in the same way, with coefficients of 0.62 and -0.55 log10 per coded unit on load pH and load conductivity. Its one significant interaction, load pH with protein load at 0.19 log10 per coded unit ($p = 0.030$), is smaller than the centre-point standard deviation of the titration and is not treated as a design constraint. The full coefficient and variance tables for XMuLV LRF and for step yield are in Appendix C.

Figure 5.1 shows the four fitted surfaces over load pH and the equilibration and wash conductivity, with the remaining factors at the centre of the design. The panel for pool HCP falls from the top left to the bottom right of the plane, and the spacing of its contours widens as load pH rises, which is the interaction seen in Table 5.6. The two viral panels rise with load pH and are flat along the conductivity axis, and the yield panel is flat in both directions.

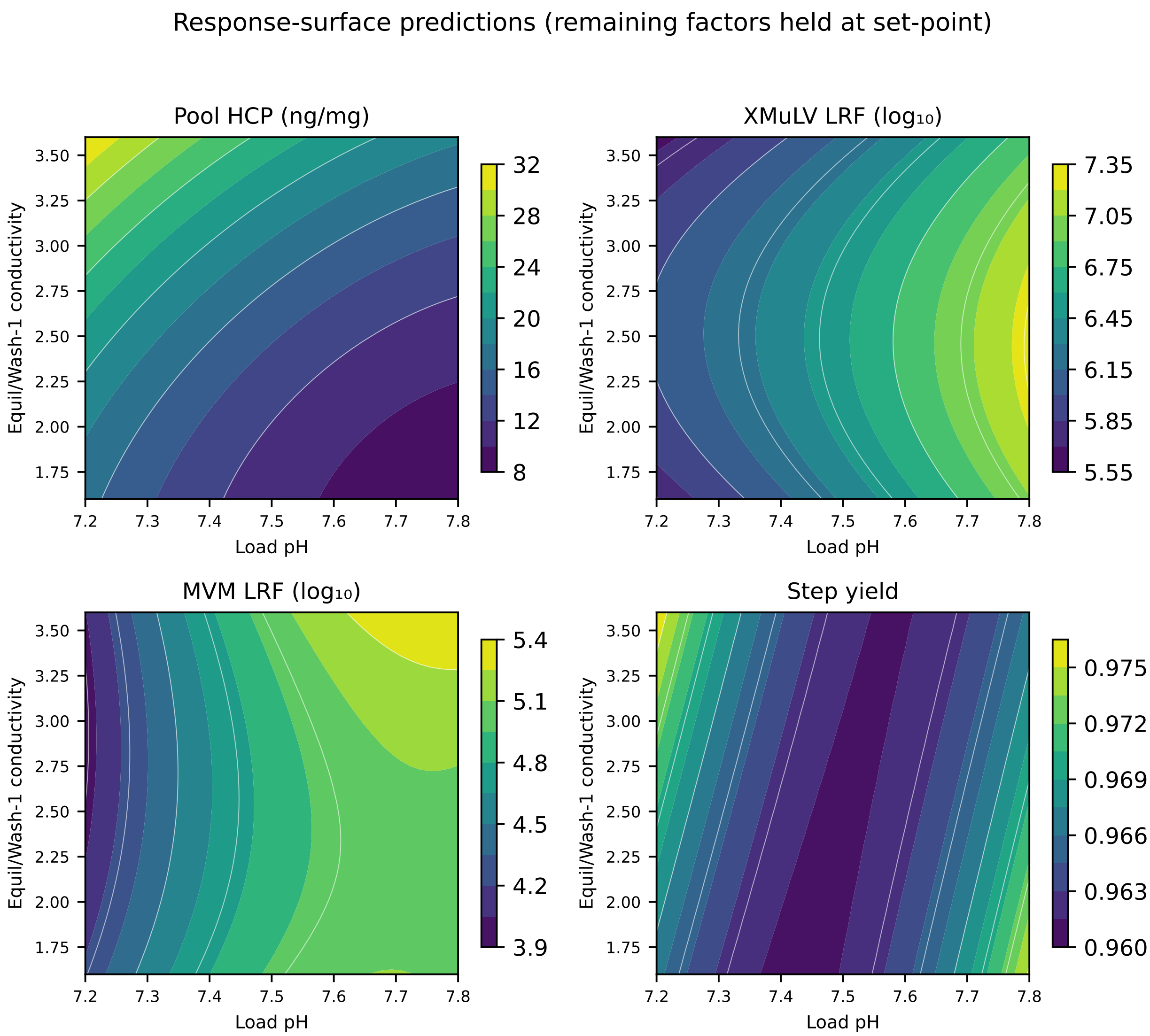

Figure 5.1: Fitted response surfaces over load pH and the equilibration and wash conductivity, remaining factors at the centre of the design.

Figure 5.2 shows the same four surfaces over load pH and load conductivity. Here both viral panels carry a gradient along both axes, with the highest clearance at high load pH with low load conductivity and the lowest at the opposite corner, and the pool HCP panel varies with load pH alone. Read together, the two figures show that each response is governed by the conductivity of one phase and not of the other, which is the point §5.4 explains.

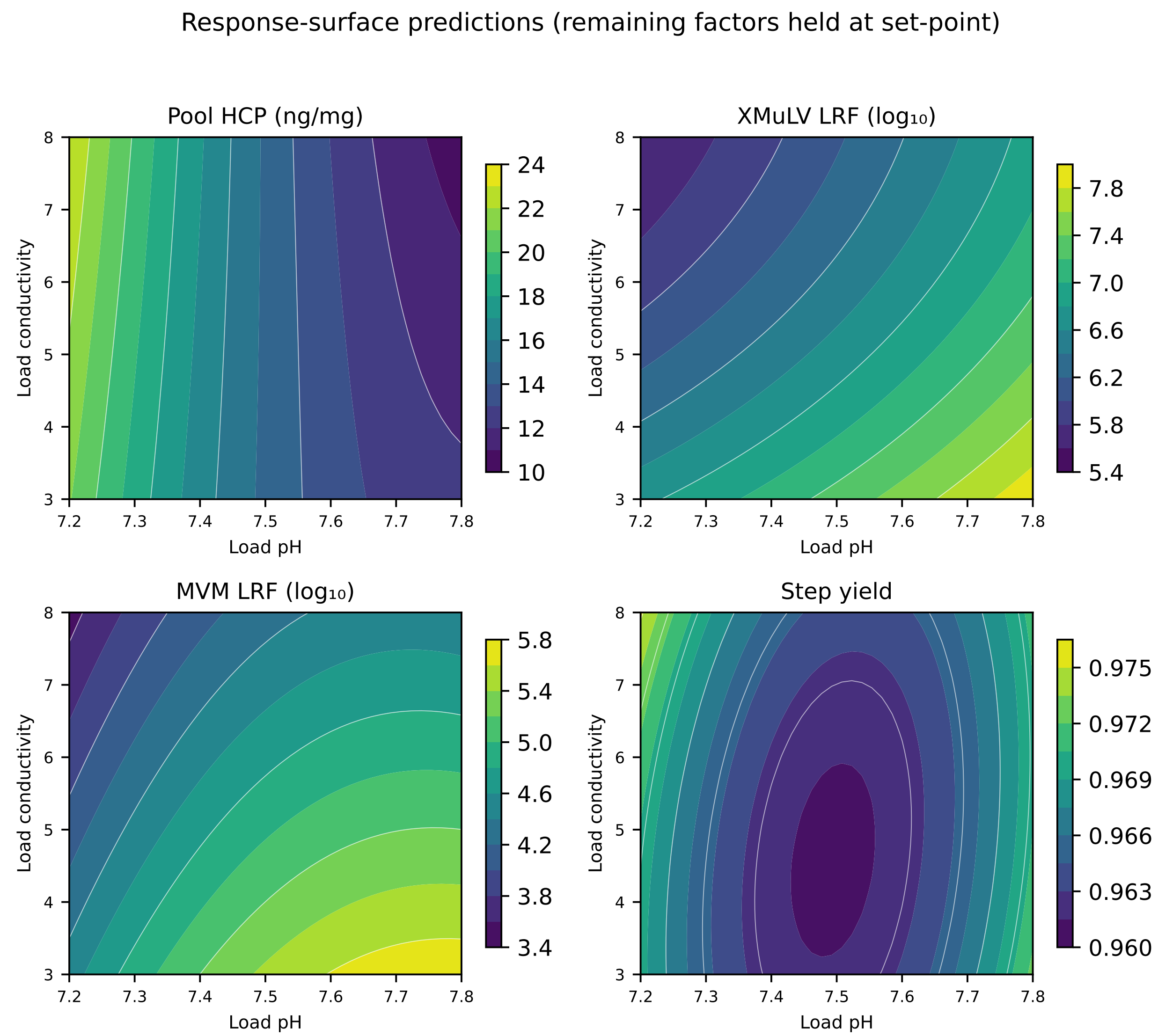

Figure 5.2: Fitted response surfaces over load pH and load conductivity, remaining factors at the centre of the design.

Figure 5.3 and Figure 5.4 give the diagnostics for the two models the control strategy rests on. In each case the standardized residuals scatter without pattern against the predicted value, the normal quantile plot is close to linear, and the actual values track the predicted values along the identity line. The diagnostics for XMuLV LRF are in Appendix C.

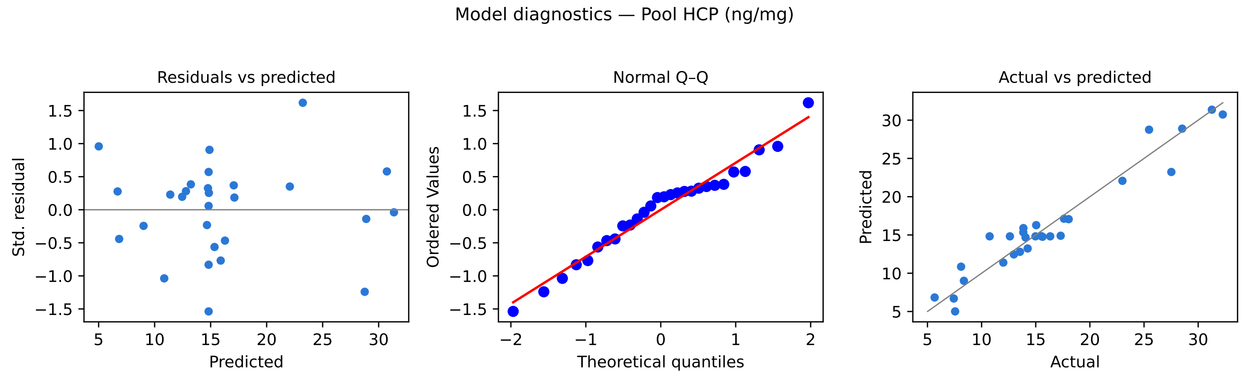

Figure 5.3: Model diagnostics for pool HCP.

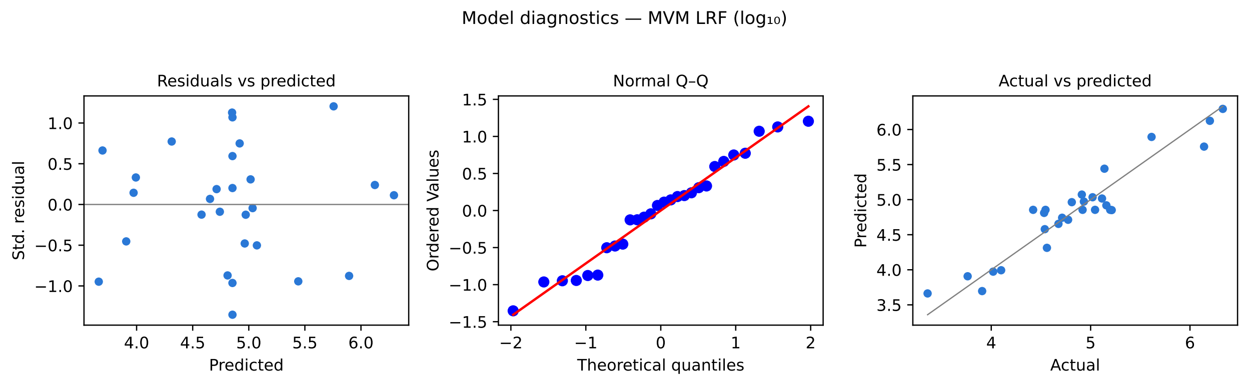

Figure 5.4: Model diagnostics for MVM LRF.

5.4 Mechanistic interpretation

The surfaces have the shape the binding equilibrium predicts, and the one result that was not anticipated is informative rather than anomalous.

Host cell protein clearance is governed by the two parameters that act on the same equilibrium from opposite sides. Raising the load pH moves more of the host cell protein population below its isoelectric point, so a larger fraction of the population is negatively charged and binds, and the pool concentration falls. Raising the conductivity of the equilibration and wash buffer raises the ionic strength in the bed, which shields the electrostatic interaction and releases the more weakly bound species into the pool. The signs of the two coefficients in Table 5.6 are opposite, which is what a single competing equilibrium requires.

The interaction between them has a direct physical reading. The slope of pool HCP against the equilibration and wash conductivity is 4.73 ng/mg per coded unit at the top of the load pH range and 7.64 ng/mg per coded unit at the bottom of it. Conductivity costs more clearance at low pH than at high pH. At low pH the bound population is dominated by species whose net negative charge is small, and a weakly held species is displaced by a given increase in ionic strength more readily than a strongly held one. Raising the load pH therefore buys robustness against conductivity as well as clearance, which is why the operating region in §6 is wider at the top of the pH range.

Virus clearance is governed by load pH in the same direction and by load conductivity rather than by the equilibration and wash conductivity. Both model viruses carry acidic surfaces and bind by the same electrostatic mechanism as the acidic host cell proteins, so the load pH dependence is expected and is nearly identical for the two viruses. The conductivity result separates the two mechanisms in time. A virus particle is a large, highly charged body present at a low number concentration, and it partitions to the ligand during the load itself, when it competes with the impurity mass arriving with the antibody. The ionic strength it experiences is therefore the ionic strength of the load. The host cell protein population, by contrast, continues to partition throughout the equilibration and the wash. The conductivity of the equilibration and wash buffer therefore sets how much of the weakly bound fraction is returned to the pool. Each response is governed by the conductivity of the phase in which its binding is decided.

Protein load carried no effect on any response, and the mechanistic reason is that the impurity mass reaching the ligand over the studied range stays far below the capacity of the bed for the bound species. The load range was widened to 300 g per litre of resin precisely to look for the point where that ceases to be true, and the surfaces show no approach to it. That conclusion is conditional on the load being representative, and §13.1 reports what happened when it was not.

Step yield is set by the collection and not by the equilibrium. The antibody remains positively charged across the whole load pH range and does not bind, so there is no mechanism by which the parameters could remove product from the pool. The measured yield of 97.3% at the nominal condition is the product of the collection boundaries and the column hold-up volume, which is why the pool collection set-point in §13.2 is the only thing that moved it.

6 Design space

The design space of the step is the region of the 4 characterized parameters over which every criterion the step governs is met. It was evaluated by predicting each of the three governed responses from its fitted response-surface model on an evenly spaced grid over the coded cube and testing every grid point against its criterion. Of the 14,641 grid points evaluated, 86 % meet every criterion simultaneously.

Pool HCP is the response that binds the region. It rejects 14 % of the characterized region on its own, and the two viral responses reject 0 % of it. The rejected region is the corner that combines low load pH with high equilibration and wash conductivity. At the corner of the characterized region least favourable to pool HCP, which sets load pH to 7.2 and the equilibration and wash conductivity to 3.6 mS/cm with load conductivity and protein load at their upper edges, the model predicts pool HCP of 30.7 ng/mg against an in-process limit of 21.7 ng/mg. That corner is an edge of failure of the step, and it lies outside the design space and far outside the normal operating ranges.

The principal plane of the design space is therefore load pH against the equilibration and wash conductivity, and it is the plane shown in Figure 5.1. Its boundary runs from low conductivity at low pH to high conductivity at high pH. The plane of load pH against load conductivity, shown in Figure 5.2, carries the viral clearance and imposes no boundary inside the characterized region, because both viral criteria are met everywhere in it. Protein load imposes no boundary on any plane.

Three ranges are nested. The characterization ranges in Table 4.1 bound the knowledge space, which is what the study explored. The design space is the part of that space in which every criterion is met, which is the 86 % just described. The normal operating ranges are the control space and are not entirely inside the design space. Predicting the three governed responses over a grid spanning the normal operating ranges, 99.6 % of that grid meets every criterion. The part that does not is the corner at which load pH sits at the bottom of its range while the equilibration and wash conductivity sits at the top of its own, where the model predicts pool HCP of 22.3 ng/mg against the in-process limit of 21.7 ng/mg. The viral responses stay well inside their criteria everywhere in the same grid, at 4.19 log10 for MVM and 5.83 log10 for XMuLV at their least favourable points. The exceedance is small and it is on an in-process limit rather than on a drug substance criterion, and §7 quantifies it parameter by parameter while §10 states the controls that respond to it.

Movement within the design space is not a regulatory change, as stated in ICH Q8(R2), but every change is still assessed under the pharmaceutical quality system of ICH Q10 (International Council for Harmonisation 2009, 2008). A planned move within the design space requires a prospective assessment against the models reported here, and a move outside it requires new data. Two bounds apply to the region as stated. It is defined on mean predictions of the fitted models, so the assurance at its boundary is approximately half, and the normal operating ranges rather than the boundary are what the control strategy commits to. It is also defined on a scale-down model and has not been confirmed at commercial scale at its edges.

7 Proven acceptable ranges

A proven acceptable range is the range of one parameter over which the attribute in question remains acceptable. Table 7.1 gives the ranges obtained for each governed attribute against each response-surface factor, in natural units, together with the criterion each was measured against and the characterization range that bounds the search.

Table 7.1: Proven acceptable ranges by attribute and parameter.

CQA	Parameter	Char. range	Unit	Crite-rion	PAR (set-point)	PAR (NOR)
Pool HCP (ng/mg)	Load pH	7.2–7.8	pH	≤ 21.7	7.21–7.8	7.45–7.8
Pool HCP (ng/mg)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	≤ 21.7	1.6–3.45	1.6–2.8
Pool HCP (ng/mg)	Load conductivity	3–8	mS/cm	≤ 21.7	3–8	3–8
Pool HCP (ng/mg)	Protein load	50–300	g/L resin	≤ 21.7	50–300	50–300
XMuLV LRF ($\log_{10}$)	Load pH	7.2–7.8	pH	≥ 3.78	7.2–7.8	7.2–7.8
XMuLV LRF ($\log_{10}$)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	≥ 3.78	1.6–3.6	1.6–3.6
XMuLV LRF ($\log_{10}$)	Load conductivity	3–8	mS/cm	≥ 3.78	3–8	3–8
XMuLV LRF ($\log_{10}$)	Protein load	50–300	g/L resin	≥ 3.78	50–300	50–300
MVM LRF ($\log_{10}$)	Load pH	7.2–7.8	pH	≥ 3.28	7.2–7.8	7.2–7.8
MVM LRF ($\log_{10}$)	Equil/Wash-1 conductivity	1.6–3.6	mS/cm	≥ 3.28	1.6–3.6	1.6–3.6
MVM LRF ($\log_{10}$)	Load conductivity	3–8	mS/cm	≥ 3.28	3–8	3–8
MVM LRF ($\log_{10}$)	Protein load	50–300	g/L resin	≥ 3.28	50–300	50–300

The criterion column needs its basis stated, because it differs between the attributes. Pool HCP is judged against an in-process limit of 21.7 ng/mg rather than against the drug substance criterion of 100 ng/mg. The in-process limit is carried back from the drug substance criterion through the clearance the steps downstream of this one deliver, and then divided by an assurance factor of 4.6. Judging an intermediate against the drug substance criterion would credit this step with clearance that later steps provide, and the assurance factor is what makes the result an in-process control rather than a break-even point. The two viral criteria are step contributions to a cumulative requirement. MVM clearance must reach 3.28 log10 at this step, which is the cumulative requirement of 8.6 log10 less the 5.32 log10 the other steps of the train deliver. XMuLV clearance must reach 3.78 log10 on the same basis.

Two analyses were run for each attribute and parameter. The first scans the fitted model over the characterization range of the parameter and reports the contiguous range in which the prediction meets the criterion. The second propagates the normal operating ranges of the other parameters by Monte-Carlo, draws the residual variability of the model with them, and reports the range in which the 95 % predictive interval stays inside the criterion. The second is a robustness statement and is narrower than the first wherever the parameters interact.

Pool HCP is the only attribute whose proven acceptable range is narrower than the characterization range. Load pH is acceptable from 7.21 upward in the first analysis and from 7.45 upward in the second, against a normal operating range that starts at 7.35. The equilibration and wash conductivity is acceptable up to 3.45 mS/cm in the first analysis and up to 2.80 mS/cm in the second, against a normal operating range that ends at 3.1 mS/cm. Load conductivity and protein load are acceptable across their whole characterization ranges for pool HCP, and all four parameters are acceptable across their whole characterization ranges for both viruses.

The two analyses do not agree about the normal operating ranges, and the difference is the result the control strategy has to answer. On the first analysis the normal operating range of every parameter lies inside its proven acceptable range. On the second the normal operating ranges of load pH and of the equilibration and wash conductivity each extend past it, at the low end for pH and at the high end for conductivity. The two analyses differ because they ask different questions. The first moves one parameter and holds the rest, and on that basis the whole of each normal operating range is acceptable. The second lets the other parameters vary within their own normal operating ranges and adds the residual variability of the model, and the upper tail of that distribution crosses the in-process limit before the parameter reaches the edge of its range. The lower part of the load pH range and the upper part of the equilibration and wash conductivity range are therefore supported for mean performance and not for the 95 % predictive bound.

Figure 7.1 shows the pool HCP range against the equilibration and wash conductivity, which is the parameter with the largest main effect on it. The left panel is the first analysis and the right panel the second, with the acceptance limit drawn, the acceptable region shaded green, and the set-point and the normal operating range marked. The widening of the predictive interval in the right panel is what moves the boundary inward.

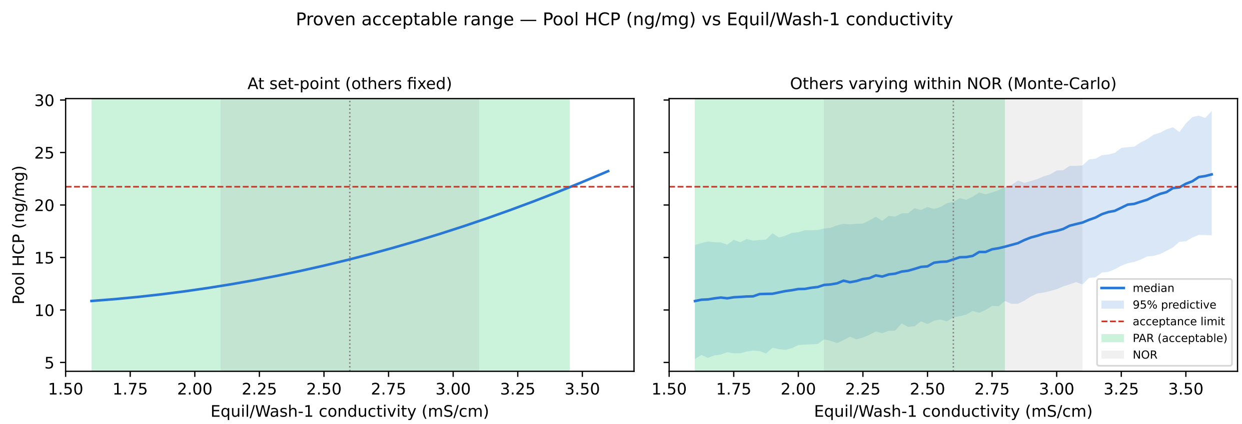

Figure 7.1: Proven acceptable range, pool HCP against the equilibration and wash conductivity.

Figure 7.2 and Figure 7.3 show the same two analyses for the two viral attributes against load pH, which is the parameter with the largest main effect on both. In each figure the predicted clearance stays above the required step contribution across the whole characterization range, including at the low pH edge, and the acceptable region covers the plot.

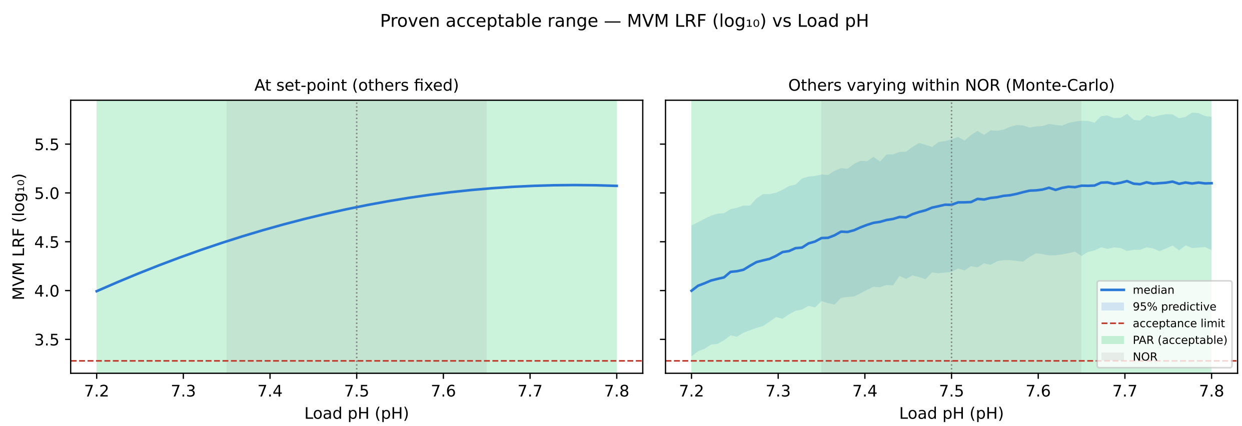

Figure 7.2: Proven acceptable range, MVM LRF against load pH.

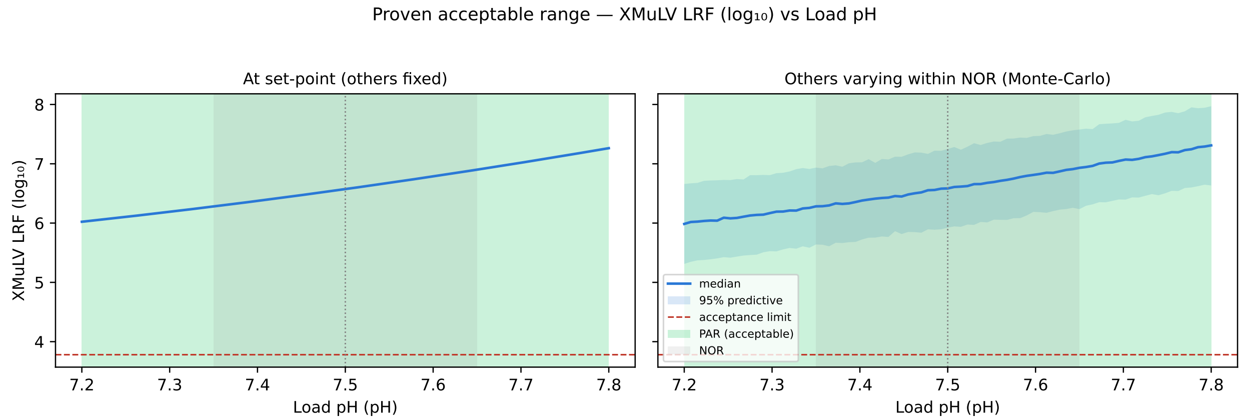

Figure 7.3: Proven acceptable range, XMuLV LRF against load pH.

The ranges in Table 7.1 are univariate statements and the design space in §6 is a multivariate one. A univariate range answers what may be done with one parameter, and it cannot express the corner in which two parameters move together. The design space is the statement that governs operation and the proven acceptable ranges are the statement that governs the assessment of a single parameter excursion. Where the two differ, as they do for load pH and the equilibration and wash conductivity, the design space is the tighter of the two and it is the one the control strategy commits to.

8 Process capability and robustness

Table 8.1 gives the commercial-scale capability of the attributes this step sets or clears. The estimate comes from a Monte-Carlo simulation of 2,000 batches with every parameter of the train varying inside its normal operating range, propagated through the fitted models of every step, and is reported as a one-sided Cpk against the drug substance criterion.

Table 8.1: Commercial-scale capability of the attributes this step sets or clears.

CQA	Crit.	Spec	Acceptance	Mean ± SD	Cpk
Host Cell Protein (HCP)	M-H	upper	0–100	15.2 ± 4.6	6.1
Residual DNA	VL	upper	0–0.001	0.0001 ± 0	10.8
Leached Protein A	L-M	upper	0–5	0.003 ± 0.0006	2,785.3
Viral clearance — XMuLV (enveloped)	VH	lower	16.7–30	19.5 ± 0.61	1.5
Viral clearance — MVM (parvovirus)	VH	lower	8.6–20	10.4 ± 0.4	1.5

Cumulative MVM clearance is the tightest capability in the table and the tightest in the drug substance, at a Cpk of 1.51 against a requirement of 8.6 log10. The simulated mean is 10.38 log10 with a standard deviation of 0.40, so the margin between the mean and the requirement is 1.78 log10. Cumulative XMuLV clearance is next at 1.53. Host cell protein, residual DNA and leached Protein A all carry capability of an entirely different order, because each is cleared by several steps in series and the drug substance criterion sits orders of magnitude above the delivered concentration.

Table 8.2 gives the modular clearance the two claims rest on. This step contributes 4.71 log10 of MVM clearance and virus filtration contributes 5.32 log10. Low-pH inactivation contributes nothing to the MVM claim, because a non-enveloped parvovirus is not inactivated by low pH. Two steps therefore carry the entire MVM claim and this is one of them, which is why its capability is the tightest of the drug substance. For XMuLV the claim rests on three steps, with 7.10 log10 from low-pH inactivation, 5.95 log10 from this step and 5.82 log10 from virus filtration. The three mechanisms are orthogonal, since low-pH treatment acts on the envelope, this step acts on surface charge and virus filtration acts on size. No clearance is claimed for Protein A or for cation exchange chromatography, although both remove virus.

Table 8.2: Modular viral clearance by step, log10 reduction factor.

step	XMuLV	MVM
Low-pH Viral Inactivation	7.1	0
Anion Exchange (AEX)	5.95	4.71
Virus Filtration	5.82	5.32
Cumulative	18.9	10

Robustness inside the normal operating ranges was assessed by predicting each governed response over the grid spanning those ranges described in §6 and taking its least favourable point. Pool HCP reaches 22.3 ng/mg, which is above its in-process limit of 21.7 ng/mg. MVM LRF falls to 4.19 log10 against the required 3.28 log10 and XMuLV LRF to 5.83 log10 against the required 3.78 log10.

The viral margins are wide and the host cell protein margin is not. The step is robust for viral clearance across its normal operating ranges by more than a log in each case, and it is at its in-process limit for host cell protein at the least favourable point in those ranges. That exceedance is bounded in two ways. It is an exceedance of an in-process limit that already carries an assurance factor of 4.6 over the concentration the downstream train can still bring to specification, and the drug substance criterion of 100 ng/mg is met with a Cpk of 6.1 at a simulated mean of 15.2 ng/mg. It is also predicted at a point the process does not operate at, since it requires load pH at the bottom of its range while the equilibration and wash conductivity sits at the top of its own. The finding is nonetheless the reason the control strategy in §10 carries both an in-process limit on the pool and a review of the conductivity range before Stage 2.

9 Parameter classification

Each parameter was classified under SOP-4001 from its demonstrated effect on a quality attribute and from the risk that it leaves the design space in routine operation. A parameter whose variability affects a critical quality attribute is a critical process parameter, and it is designated well controlled where the risk of falling outside the design space is low (CMC Biotech Working Group 2009). Table 4.1 carries the outcome.

- Load pH was classified as a well controlled critical process parameter. It carries the largest effect on host cell protein clearance and the largest effect on both viral clearances, and it is the parameter that sets the position of the binding equilibrium for every bound species. It is designated well controlled because it is set by titration of a defined buffer system before the load and is measured in process, and because the proven acceptable range extends beyond the normal operating range in the direction that matters.
- The equilibration and wash conductivity was classified as a well controlled critical process parameter. It carries the largest effect on pool host cell protein and it is the only parameter whose normal operating range extends past the range the robustness analysis supports (§7). It is designated well controlled rather than critical because buffer conductivity is verified at release and monitored in line during equilibration and washing, so the control capability of the equipment is far tighter than the range, and because the attribute it acts on is held by an in-process limit as well as by the range.
- Load conductivity was classified as a well controlled critical process parameter. It carries no effect on host cell protein clearance and a significant effect on the clearance of both model viruses, at -0.56 log10 per coded unit for MVM. It is designated well controlled because the load conductivity is set by the adjustment of the cation exchange pool and is verified before the load begins.
- Protein load was classified as a well controlled critical process parameter. No effect on any response was demonstrated over 50 to 300 g per litre of resin, and the mechanistic reason is that the impurity mass stays far below the capacity of the bed (§5.4). It is retained in the critical class rather than reduced to a key process parameter because it is the parameter through which a change in load composition would express itself, which is what DEV-008-01 demonstrated (§13.1).
- Operating flow rate was classified as a well controlled critical process parameter. It was assessed one parameter at a time (§4.4) and it acts on the clearance of every bound species through residence time, which is a quality-linked path even though the studied range showed no consequence. It is designated well controlled

because flow is set and recorded by the chromatography skid and its range is wide relative to the control capability of the equipment.

All 5 parameters are therefore well controlled critical process parameters and the step has no critical process parameter requiring a narrower control than the equipment already provides. It also has no key process parameter and no general process parameter, which follows from the step governing a very high criticality attribute through parameters that all act on one equilibrium.

10 Contribution to the control strategy

The step contributes five elements to the control strategy of the drug substance process.

The first is the set of parameter ranges in Table 4.1. Each parameter is controlled to its normal operating range. Those ranges lie inside the design space defined in §6 except at the corner combining the bottom of the load pH range with the top of the equilibration and wash conductivity range, and the robustness analysis in §7 places the supported region at load pH at or above 7.45 and the equilibration and wash conductivity at or below 2.8 mS/cm. Two actions follow. The normal operating range of the equilibration and wash conductivity should be reviewed against Table 7.1 before the Stage 2 ranges are fixed, and until it is, operation at the top of that range should be avoided when the load pH is at the bottom of its own. Neither parameter needs a tighter control than the equipment already delivers, which is why both remain well controlled critical process parameters (§9).

The second is in-process control of pool host cell protein against the limit of 21.7 ng/mg derived in §7. The nominal step delivers 18.9 ng/mg, and the limit is what confirms that the step handed the downstream train a load it can bring to the drug substance criterion. This control is what catches a batch that runs at the unfavourable corner described above, and it is the reason that corner is a monitored risk rather than an undetected one. Residual DNA and leached Protein A are monitored across the step and are controlled by release testing of the drug substance rather than by an in-process limit here, because their clearance is robust to the operating parameters.

The third is the modular viral clearance claim of §8. The step contributes 4.71 log10 of MVM clearance and 5.95 log10 of XMuLV clearance to a cumulative claim consolidated in PCMR-001. Load pH and load conductivity are the two parameters that claim depends on, and both are controlled and verified before the load begins.

The fourth is a control on the feed to the step, which arises from DEV-008-01. A maximum hold time was placed on the neutralized cation exchange pool before it is adjusted and loaded, and charge variant content of the load is verified by icIEF under AMV-3013 before the step is run. Neither control existed when the first execution of the designs was run and the deviation is the reason both exist now (§13.1).

The fifth is life-cycle control of the resin under SOP-2008. The models reported here were fitted on a resin operated within its qualified cycle limits, and the clearance the step claims is conditional on that. Cycle number, sanitization and column efficiency are therefore controlled and trended, and column repacking is triggered on efficiency rather than on cycle count alone.

Attributes this step does not control are named here so that the boundary of the claim is explicit. The glycan attributes, the charge variants and the aggregate content of the drug substance are formed or reduced upstream and are reported in PCR-003 and PCR-007. Leached Protein A is set by the capture step and is reported in PCR-005. Enveloped virus clearance is set principally by low-pH inactivation and is reported in PCR-006. The remaining MVM clearance is delivered by virus filtration and is reported in PCR-009. This report claims only what Table 8.1 and Table 8.2 attribute to Step 8.

11 Discussion

The study establishes that the anion exchange step is governed by two parameters and that they act through one binding equilibrium. Load pH sets the net charge of every species in the feed relative to its isoelectric point, and conductivity sets how strongly that charge is shielded. What the study adds to prior knowledge is the separation between the two conductivities. Host cell protein clearance is set by the conductivity of the equilibration and wash buffer, and viral clearance is set by the conductivity of the load. The two attributes are therefore controlled by different parameters, and an excursion in one conductivity does not carry an implication for the other attribute. That is a result the platform history did not contain, and it is the reason the step can hold a tight host cell protein limit and a wide viral margin at the same time.

Confidence in the scale-down model rests on three things. The model matched at-scale performance on the attributes that matter to this step at the set-point condition (§3.1). Its centre-point replicates reproduce at 18.2 % on the least precise response, which is the precision of the assay rather than of the process. Lack of fit is not significant for any of the three governed responses, which means the models describe the data to within the reproducibility the replicates establish. The viral results carry an additional qualification, since they were generated on a separate spiking model whose load is spiked with live virus. That model reproduces the process model's geometry, resin and buffers, but the spike itself is a difference from the process and the clearance claim is bounded by it.

Three deviations were recorded and are reported in §13. One of them invalidated the first execution of both designs. The load used in that execution had been held neutralized long enough to raise its acidic charge variant content by roughly half, and the deamidated antibody competed with the impurities for the ligand at high load. Both designs were re-executed in full on a requalified load and the anomalous interaction is absent from the re-executed data. The other two deviations were bounded and did not change any conclusion.

One result of the study is uncomfortable and is stated here rather than left in the tables. The normal operating ranges as they stand are not fully supported. The corner combining the bottom of the load pH range with the top of the equilibration and wash conductivity range puts the predicted pool host cell protein at 22.3 ng/mg against an in-process limit of 21.7 ng/mg, and the robustness analysis withdraws support from the lower part of the load pH range and the upper part of the conductivity range. The consequence for the drug substance is small, because the in-process limit carries an assurance factor of 4.6 and the drug substance criterion is met with wide margin (§8). The consequence for the control strategy is not small, and §10 carries it as an action rather than as an observation.

Four limitations bound what this report claims. The models are fitted on a scale-down model, and the design space has not been confirmed at commercial scale at its edges. The models predict mean levels, so the assurance at the edge of the design space is approximately half and the control strategy commits to the normal operating ranges rather than to the boundary. The viral clearance claim is modular and rests on a small scale spiking model, so it is a claim about the step and not about the process, and the cumulative claim is made in PCMR-001. The clearance the step delivers is conditional on the resin being operated inside its qualified life cycle, which is a control rather than a result of this study. Each of these is managed forward by the elements listed in §10 and each is verified in Stage 2.

12 Conclusions

The anion exchange step has been characterized on a qualified scale-down model over ranges at least twice the width of the intended control ranges. Load pH and the equilibration and wash conductivity govern host cell protein clearance and interact. Load pH and load conductivity govern the clearance of both model viruses. Protein load carries no demonstrated effect over the range studied and the univariate assessment of operating flow rate identified none over its own. A design space has been defined over 86 % of the characterized region, bounded by pool host cell protein at the corner combining low load pH with high equilibration and wash conductivity. The normal operating ranges lie inside that region other than at the same corner, where the predicted pool host cell protein reaches 22.3 ng/mg against an in-process limit of 21.7 ng/mg, and the range of the equilibration and wash conductivity is to be reviewed against the proven acceptable ranges before the Stage 2 ranges are fixed. All 5 parameters were classified as well controlled critical process parameters.

The attributes the step governs meet their criteria at commercial scale. Cumulative MVM clearance carries a Cpk of 1.51, which is the tightest capability of the drug substance, and the step contributes 4.71 log10 of that clearance. Three deviations were recorded, investigated and dispositioned, and the one that invalidated the first execution of the designs was resolved by re-executing both designs in full on a re-qualified load. The step is ready for Stage 2 verification and its outcomes roll up into PCMR-001.

13 Deviations from the plan

Execution of the study recorded 3 deviations from PCP-008. Table 13.1 lists them with the point at which each was detected and its disposition. Each is then reported with what went wrong, what the investigation found, what the impact was and how it was dispositioned.

Table 13.1: Deviations recorded during execution of PCP-008.

Deviation	Summary	Detected during	Disposition
DEV-008-01	Non-representative (deamidated) load in the first DoE execution; designs invalidated and re-executed	load-material charge-variant verification by icIEF (AMV-3013)	invalidated_and_re_executed
DEV-008-02	Trailing-edge UV pool-collection set-point slightly permissive in both executions	post-execution pool-fraction reconciliation (AMV-3012)	corrected_by_modelling_and_verification_runs
DEV-008-03	Equilibration/Wash-1 buffer released below its pH acceptance range on one screening run	buffer release testing per SOP-2103	retained

13.1 DEV-008-01, non-representative load in the first execution

In the first execution of both designs the flow-through pool carried up to 150 ng/mg host cell protein, against an in-process limit of 21.7 ng/mg and a mean across the design of 32 ng/mg. The high results were confined to the runs that combined the highest protein load with the highest equilibration and wash conductivity, and that pattern appeared in the screening design as an interaction between protein load and the equilibration and wash conductivity of 34.8 ng/mg ($p = 0.002$), which was the largest interaction in that dataset. Prior knowledge offered no mechanism by which protein load should interact with that conductivity at loads far below the capacity of the bed.

The investigation examined the load material. Charge variant analysis of the load lot by icIEF under AMV-3013 returned acidic charge variants at 33.5 % of total, against 21.0 % in representative material. The batch record for the cation exchange pool lot

LOT-CEX-8842 showed a neutralized hold of 36 hours before the pool was adjusted and loaded. The root cause was assigned as asparagine deamidation promoted by that extended neutralized hold. Deamidation converts an amide to a carboxylate, so it lowers the isoelectric point of the affected molecules. A fraction of the antibody therefore carried enough negative charge at the load pH to bind the ligand, compete with the impurities for capacity and displace weakly bound host cell protein into the pool. The interaction with protein load follows directly, because the competing product mass rises with the load.

The impact assessment concluded that the load was not representative of the material the commercial process delivers to the step, and that the operating region derived from it would be a region for a deamidated feed rather than for the process. Both designs were therefore invalidated for operating region definition and were re-executed in full on a requalified, representative cation exchange pool. Root cause was confirmed from the re-executed data. In the requalified screening design the same interaction is 0.22 ng/mg ($p = 0.860$), which is indistinguishable from zero, and the highest pool host cell protein anywhere in the design falls to 36 ng/mg with a design mean of 17 ng/mg. The anomaly is a property of the load and not of the step.

The first execution is retained as a superseded dataset. It was used only to confirm root cause and no result in §5 to §9 of this report comes from it. Two controls were added to the step as a result, a maximum hold time on the neutralized cation exchange pool and verification of the load charge variant content by icIEF before the step is run (§10).

13.2 DEV-008-02, permissive pool collection set-point

Collection of the flow-through pool was stopped on the trailing edge of the ultraviolet trace at 180 mAU in both executions of the designs. Reconciliation of the pool fractions after execution showed that this set-point collects further into the trailing edge than the method intends, and the corrected set-point is 120 mAU. The trailing edge is the part of the flow-through that carries the highest host cell protein concentration, since weakly bound species are released late.

The impact is a bias on one response and it acts in one direction. Collecting further into the trailing edge adds host cell protein to the pool, so every pool host cell protein result in this report is biased high relative to the corrected method, and no result is biased low. The bias is common to every run of both designs, since the set-point was the same on all of them, so it displaces the fitted surface without changing the effects estimated from it. The in-process limit, the design space boundary and the proven acceptable ranges derived in §6 and §7 are therefore conservative with respect to the corrected method.

The correction was made in the method and was verified. 3 verification runs were executed at the corrected set-point, of which 2 were at corners of the normal operating ranges. Pool host cell protein at the corrected set-point was consistent with the fitted surface after allowing for the removed trailing edge. The verification set is small and its mean carries a 95 % confidence interval of 23.6 % relative, derived from the precision of AMV-3012 and the number of runs, which is why the verification is reported as confirming the direction and magnitude of the correction rather than as a re-estimate of the model. The deviation was dispositioned as corrected by modelling and verification runs, and the corrected set-point is what the commercial method specifies.

13.3 DEV-008-03, equilibration and wash buffer released below its pH range

The equilibration and wash buffer lot LOT-BUF-8815 was released at pH 7.32 against an acceptance range whose lower limit is 7.4 and a target of 7.5. The lot was used on one screening run. The deviation was detected at buffer release testing under SOP-2103 and the root cause was assigned as a titration endpoint error at buffer preparation.

The impact was bounded from the effects measured in this study. A lower buffer pH weakens the binding of the acidic species in the bed, which acts in the same direction as a lower load pH. Taking the load pH coefficient of the response-surface model as an upper bound on the sensitivity of the step to a pH offset of this size, an offset of 0.18 pH units corresponds to at most 3.2 ng/mg on the pool host cell protein of that run and at most 0.32 log10 on its MVM clearance. The host cell protein bound is below the centre-point standard deviation of the assay reported in §5.1 and the clearance bound is within the reproducibility of the titration. The affected run is a factorial run of the screening design, and the screening design is used in §5.2 to identify which factors carry effects rather than to predict.

The run was therefore retained in the analysis and the deviation was dispositioned as retained. The buffer preparation procedure was reviewed and the titration endpoint check was reinforced under SOP-2103. No result in this report depends on that single run.

Appendix A. Screening design matrix and responses

Table 13.2 gives the coded settings of the screening design and Table 13.3 the measured responses of each run. The coded letters are those of Table 4.2.

Table 13.2: Screening design, coded factor settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

Table 13.3: Screening design, measured responses.

Run	Pool HCP (ng/mg)	XMuLV LRF ($\log_{10}$)	MVM LRF ($\log_{10}$)	Step yield
1	12.5	6.16	4.59	0.99
2	15.6	6.15	4.43	0.975
3	12.1	5.47	3.76	0.951
4	14.6	5.56	3.96	0.964
5	35.7	6.34	4.74	0.965
6	35	6.3	4.83	0.995
7	29.2	5.92	4.01	0.974

Run	Pool HCP (ng/mg)	XMuLV LRF (log_{10})	MVM LRF (log_{10})	Step yield
8	29.8	5.95	3.94	0.969
9	9.62	7	5.91	0.973
10	7.37	7.4	5.61	0.953
11	8.69	6.61	5.25	0.972
12	8.04	6.71	4.89	0.973
13	12.7	6.59	5.91	0.995
14	15	8.34	5.97	0.981
15	12.9	6.69	5.15	0.966
16	15	6.52	5.31	0.965
17	16.2	6.42	4.54	0.966
18	17.5	6.8	5.08	0.975
19	11.6	5.95	4.83	0.974

Appendix B. Response-surface design matrix and responses

Table 13.4 gives the coded settings of the face-centred central composite design and Table 13.5 the measured responses of each run.

Table 13.4: Response-surface design, coded factor settings.

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

Table 13.5: Response-surface design, measured responses.

Run	Pool HCP (ng/mg)	XMuLV LRF (log_{10})	MVM LRF (log_{10})	Step yield
1	13.5	6.72	4.55	0.951
2	17.3	6.03	5.21	0.989
3	14.1	5.87	4.02	0.995
4	15	5.3	3.76	0.951
5	28.5	6.53	4.77	0.988
6	25.5	6.71	4.71	0.963
7	31.3	5.36	3.91	0.995
8	32.3	5.06	3.36	0.966
9	5.66	7.5	6.14	0.974
10	8.37	7.83	5.61	0.989
11	7.56	6.49	4.54	0.972
12	7.43	7.13	4.68	0.988
13	17.6	7.18	6.2	0.974
14	18	8.12	6.33	0.968
15	13.9	6.58	4.53	0.96
16	13.9	6.37	5.16	0.973
17	23	5.83	4.1	0.971
18	12	7.37	4.91	0.965
19	8.1	6.42	4.93	0.946
20	27.5	6	5.02	0.972
21	15.6	7.44	5.14	0.952
22	15.5	5.93	4.56	0.967
23	13	6.52	4.81	0.968
24	14.3	6.7	5.11	0.952
25	15	6.59	4.42	0.958
26	10.7	6.03	5.2	0.975
27	16.3	7.05	5.04	0.977
28	12.6	6.86	4.92	0.953

Appendix C. Full effect and coefficient tables

Table 13.6 to Table 13.9 give every estimated effect of the screening models, including the terms not significant at $\alpha = 0.05$.

Table 13.6: Screening effects, pool HCP, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
B	12.1	6.04	0.613	9.85	9.48e-06	***
A	-11.9	-5.96	0.613	-9.72	1.05e-05	***
AB	-6.63	-3.32	0.613	-5.41	0.000642	***
C	-1.66	-0.829	0.613	-1.35	0.213	
AC	1.65	0.827	0.613	1.35	0.214	
BC	-1.21	-0.605	0.613	-0.986	0.353	
D	0.885	0.443	0.613	0.722	0.491	
AD	-0.498	-0.249	0.613	-0.406	0.696	
CD	0.261	0.13	0.613	0.212	0.837	
BD	0.224	0.112	0.613	0.182	0.86	

Table 13.7: Screening effects, XMuLV LRF, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1	0.501	0.0982	5.1	0.000934	***
C	-0.605	-0.302	0.0982	-3.08	0.0151	*
D	0.268	0.134	0.0982	1.37	0.209	
CD	-0.255	-0.128	0.0982	-1.3	0.23	
AD	0.25	0.125	0.0982	1.27	0.24	
B	0.198	0.0991	0.0982	1.01	0.342	
BD	0.126	0.0632	0.0982	0.643	0.538	
AC	-0.0968	-0.0484	0.0982	-0.493	0.635	
AB	-0.094	-0.047	0.0982	-0.478	0.645	
BC	-0.0169	-0.00844	0.0982	-0.0859	0.934	

Table 13.8: Screening effects, MVM LRF, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
A	1.22	0.61	0.0446	13.7	7.83e-07	***
C	-0.715	-0.357	0.0446	-8.01	4.32e-05	***
B	0.182	0.091	0.0446	2.04	0.0757	
BD	0.107	0.0536	0.0446	1.2	0.263	
AD	-0.0627	-0.0314	0.0446	-0.704	0.502	
D	-0.0451	-0.0226	0.0446	-0.506	0.626	
BC	-0.0442	-0.0221	0.0446	-0.496	0.634	
CD	0.0334	0.0167	0.0446	0.375	0.718	
AC	0.0135	0.00677	0.0446	0.152	0.883	
AB	-0.0132	-0.00659	0.0446	-0.148	0.886	

Table 13.9: Screening effects, step yield, all terms.

Term	Effect	Coef.	Std. err.	t	p-value	Sig.
C	-0.0116	-0.00582	0.00315	-1.85	0.102	
B	0.00743	0.00371	0.00315	1.18	0.272	
AD	-0.00715	-0.00357	0.00315	-1.13	0.29	
AC	0.00513	0.00257	0.00315	0.815	0.439	
BC	-0.00392	-0.00196	0.00315	-0.622	0.551	
BD	0.00383	0.00192	0.00315	0.608	0.56	
CD	0.00346	0.00173	0.00315	0.549	0.598	
AB	0.00141	0.000707	0.00315	0.224	0.828	
D	-0.00136	-0.000679	0.00315	-0.215	0.835	
A	-0.000663	-0.000332	0.00315	-0.105	0.919	

Table 13.10 to Table 13.13 give the response-surface coefficients and the analysis of variance for the two responses not shown in §5.3.

Table 13.10: Response-surface coefficients, XMuLV LRF.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	6.57	0.109	60.6	2.51e-17	***
A	0.621	0.074	8.39	1.33e-06	***
B	-0.076	0.074	-1.03	0.323	
C	-0.554	0.074	-7.48	4.6e-06	***
D	0.0287	0.074	0.388	0.704	
AB	-0.0278	0.0785	-0.355	0.729	
AC	0.0223	0.0785	0.285	0.78	
AD	0.192	0.0785	2.44	0.0295	*
BC	-0.118	0.0785	-1.5	0.158	
BD	0.0577	0.0785	0.735	0.475	

Term	Coef.	Std. err.	t	p-value	Sig.
CD	-0.0751	0.0785	-0.957	0.356	
A^2	0.0696	0.195	0.356	0.727	
B^2	-0.328	0.195	-1.68	0.118	
C^2	0.152	0.195	0.779	0.45	
D^2	0.0761	0.195	0.39	0.703	

Table 13.11: Analysis of variance with lack of fit, XMuLV LRF.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	13.8	14	0.988	10	8.69e-05
Residual	1.28	13	0.0985	nan	nan
Lack of fit	0.687	10	0.0687	0.347	0.911
Pure error	0.594	3	0.198	nan	nan

Table 13.12: Response-surface coefficients, step yield.

Term	Coef.	Std. err.	t	p-value	Sig.
Intercept	0.961	0.00501	192	7.92e-24	***
A	-0.000294	0.00342	-0.086	0.933	
B	0.00022	0.00342	0.0643	0.95	
C	0.00109	0.00342	0.318	0.756	
D	-0.00203	0.00342	-0.594	0.563	
AB	-0.00464	0.00362	-1.28	0.223	
AC	-0.00175	0.00362	-0.482	0.637	
AD	0.00608	0.00362	1.68	0.117	
BC	-0.000152	0.00362	-0.042	0.967	
BD	-0.00433	0.00362	-1.2	0.253	
CD	-0.00413	0.00362	-1.14	0.275	
A^2	0.00932	0.00902	1.03	0.32	
B^2	0.000524	0.00902	0.0581	0.955	
C^2	0.00152	0.00902	0.168	0.869	
D^2	0.00177	0.00902	0.196	0.847	

Table 13.13: Analysis of variance with lack of fit, step yield.

Source	Sum sq.	df	Mean sq.	F	p-value
Model	0.0026	14	0.000186	0.885	0.589
Residual	0.00273	13	0.00021	nan	nan
Lack of fit	0.00228	10	0.000228	1.51	0.407
Pure error	0.000453	3	0.000151	nan	nan

Figure 13.1 gives the diagnostics of the XMuLV model.

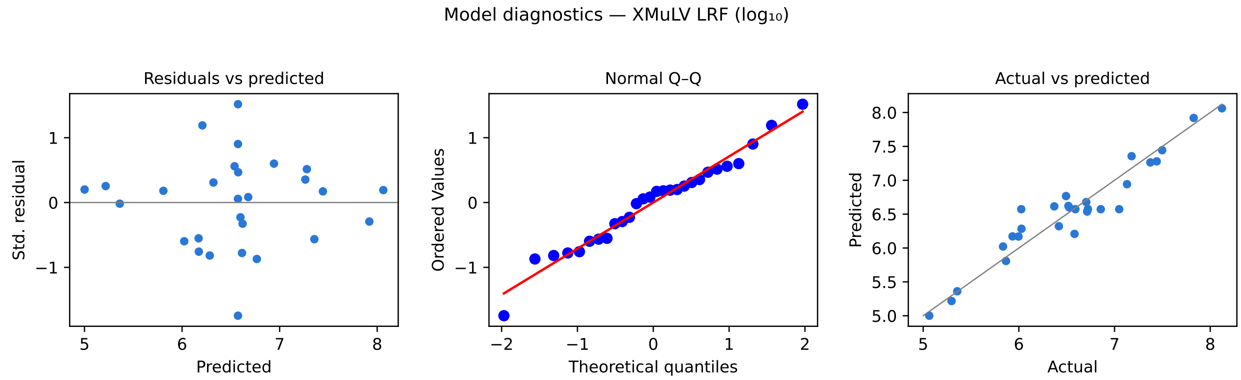

Figure 13.1: Model diagnostics for XMuLV LRF.

Table 13.14 gives the centre-point reproducibility of the screening design, which supports the comparison made in §5.1.

Table 13.14: Centre-point reproducibility, screening design.

Response	n	Mean	SD	%CV
Pool HCP (ng/mg)	3	15.1	3.11	20.6
XMuLV LRF ($\log_{10}$)	3	6.39	0.426	6.67
MVM LRF ($\log_{10}$)	3	4.81	0.268	5.57
Step yield	3	0.972	0.00523	0.538

Appendix D. Analytical methods summary

Table 13.15 lists the validated methods used in the study with the analyte each measures and the form in which the result is reported.

Table 13.15: Validated analytical methods used in the study.

Method	Title	Analyte	Technique	Reportable
AMV-3012	Host-Cell Protein ELISA	Host cell protein	Immunoassay	ng/mg
AMV-3014	Residual DNA (qPCR)	Residual DNA	Quantitative PCR	ng/dose
AMV-3016	Leached Protein A by ELISA (ppm)	Leached Protein A	Immunoassay	ppm
AMV-3017	XMuLV Infectivity Titre (TCID50)	XMuLV infectivity	Cell-based titration	log10 LRF
AMV-3018	MVM Infectivity Titre (TCID50/qPCR)	MVM infectivity	Cell-based titration and qPCR	log10 LRF
AMV-3013	Charge Variants (icIEF)	Charge variants	Imaged capillary isoelectric focusing	% of total

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